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lab task 1 lab 6.cpp
1  #include <iostream>
2  #include <fstream>
3  #include <string>
4  using namespace std;
5  struct sensor_reading
6  {
7      int sensorID;
8      float temperature;
9      float humidity;
10     string timestamp;
11 };
12 int main()
13 {
14     sensor_reading sensors[3] =
15     {
16         {1, 36.5, 72.3, "2024-01-15 08:00"},
17         {2, 38.1, 68.9, "2024-01-15 08:05"},
18         {3, 35.7, 75.1, "2024-01-15 08:10"}
19     };
20     ofstream outFile("sensor_log.txt");
21     if (!outFile)
22     {
23         cout << "ERROR opening file for writing!" << endl;
24         return 1;
25     }
26     for (int i = 0; i < 3; i++)
27     {
28         outFile << "--- Sensor Reading " << i + 1 << " ---" << endl;
29         outFile << "Sensor ID : " << sensors[i].sensorID << endl;
30         outFile << "Temperature : " << sensors[i].temperature << " C" << endl;
31         outFile << "Humidity : " << sensors[i].humidity << " %" << endl;
32         outFile << "Timestamp : " << sensors[i].timestamp << endl;
33         outFile << "-----" << endl;
34     }
35     outFile.close();
36     cout << "File \"sensor_log.txt\" created successfully." << endl;
37     ifstream inFile("sensor_log.txt");
38     if (!inFile)
39     {
40         cout << "ERROR opening file for reading!" << endl;
41         return 1;
42     }
43     cout << "\nReading file contents:\n" << endl;
44     string line;

```

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C:\Users\laha's Laptop\OneDrive\Desktop\assignment 2 coop\lab task 1 lab 6.exe
File "sensor_log.txt" created successfully.
Reading file contents:
--- Sensor Reading 1 ---
Sensor ID : 1
Temperature : 36.5 C
Humidity : 72.3 %
Timestamp : 2024-01-15 08:00
-----
--- Sensor Reading 2 ---
Sensor ID : 2
Temperature : 38.1 C
Humidity : 68.9 %
Timestamp : 2024-01-15 08:05
-----
--- Sensor Reading 3 ---
Sensor ID : 3
Temperature : 35.7 C
Humidity : 75.1 %
Timestamp : 2024-01-15 08:10
-----
-----
Process exited after 0.5099 seconds with return value 0
Press any key to continue . . .

```